

Capstone Project Report

By: Tanay Ayiliath

TRIPLE POWER LAW FIT ANALYSIS RESULTS

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DATA STATISTICS:

Total data points: 628

logMacc range: -10.469 to -4.777

logAge range: 5.010 to 7.780

Mstar range: 0.149 to 10.037 Solar Masses

CORRELATIONS:

logAge vs logMacc: -0.150

logMstar vs logMacc: 0.537

logAge vs logMstar: 0.381

TRIPLE POWER LAW FIT RESULTS:

$\log(\text{Macc}) = -4.359 + -0.585 \times \log(\text{Age}) + 1.875 \times \log(\text{Mstar})$

$\text{Macc} = 4.372\text{e-}05 \times \text{Age}^{-0.585} \times \text{Mstar}^{1.875}$

INTERPRETATION:

→ Mass accretion decreases with age (expected)

→ More massive stars have higher accretion rates (expected)

Mass Accretion Rate vs Age: Triple Power Law Fit Colored by Stellar Mass

